HR Analytics

**Project Title**

# Workforce DNA: Decoding Employee Behavior and Organizational Trends

Python (Google Colab) & Power BI Desktop

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## Project Summary:

- Analyse HR employee data to understand employee attrition and workforce trends across different departments and job roles.

- Use Python for data cleaning, analysis, and visualization to find insights related to age, gender, education, job satisfaction, and employee attrition.

- Develop an interactive Power BI dashboard with KPIs, charts, and slicers to present HR insights clearly and help organizations make better employee retention decisions.

## Dataset Description

- The project uses the **IBM HR Analytics Employee Attrition & Performance** dataset, which contains **1,470 employee records** and **23 attributes**. The dataset includes employee demographic information, job details, salary, work experience, job satisfaction, overtime, and attrition status. It was used to analyze employee attrition trends and generate business insights using Python and Power BI.

## Tools & Software Used

- **Python (Google Colab)** – Data Cleaning, Data Pre-processing, Exploratory Data Analysis (EDA), and Data Manipulation.

- **NumPy** – Numerical calculations and array-based operations.

- **Pandas** – Data loading, cleaning, transformation, grouping, filtering, and manipulation.

- **Matplotlib** – Created statistical charts and graphical visualizations.

- **Seaborn** – Developed advanced visualizations such as heatmaps, box plots, violin plots, and pair plots.

- **Power BI Desktop** – Designed interactive dashboards with KPI cards, charts, slicers, and business insights

## Python (Google Colab)

## Data Collection

- Collected the **IBM HR Analytics Employee Attrition Dataset** from GitHub containing employee demographic details, job information, education, salary, work experience, satisfaction ratings, performance ratings, and attrition status.
- The dataset consists of **1,470 employee records and 23 columns**.
- Imported the dataset into Python using the **Pandas** library for further analysis.
- Verified the dataset by checking the number of rows, columns, data types, duplicate records, and missing values before performing Exploratory Data Analysis.

## Data Cleaning

- Checked the dataset for duplicate records and confirmed that no duplicate values were present.
- Identified missing values in categorical and numerical columns.
- Replaced missing values in the **Monthly Income** column using the **Mean** value.
- Filled missing values in the **Education** and **Marital Status** columns with **"Unknown"**.
- Removed unnecessary white spaces from column names to maintain consistency.
- Verified the dataset after cleaning to ensure that no missing values remained.

## Data Manipulation

- Created a new **Income Level** column by categorizing employees into **Low**, **Medium**, and **High** income groups based on Monthly Income.
- Generated an **Age Group** column to classify employees into different age categories such as **18–25, 26–35, 36–45, 46–55, and 56–65**.
- Created an **Age Band** column to simplify dashboard visualization and employee segmentation.
- Filtered employees based on **Attrition Status** to analyze employees who left the organization.
- Filtered employees working **Over Time** to study its impact on employee attrition.
- Performed aggregation techniques to calculate **Average Age by Gender** and **Average Monthly Income by Department**.
- Prepared the cleaned dataset and exported it as a **CSV file** for Power BI dashboard development

## Visualization:

**1.Histogram – Monthly Income Distribution**

- The histogram illustrates the distribution of employee monthly income across the organization. Most employees earn salaries between the low and medium income ranges, while only a small number receive higher salaries. The income distribution is positively skewed, indicating fewer employees in the highest salary category. This analysis helps HR understand the overall salary structure within the organization.

**2. Bar Chart – Employee Distribution by Age Group**

- The bar chart represents the number of employees in different age groups. The majority of employees belong to the **26–35** and **36–45** age categories, indicating a workforce dominated by mid-career professionals. Younger and older employees are comparatively fewer in number. This helps HR understand workforce demographics and future succession planning.

**3. Pie Chart – Employee Attrition Distribution**

- The pie chart shows the proportion of employees who stayed and those who left the organization. Around **84%** of employees continue working, while **16%** have left the company. This indicates a relatively healthy employee retention rate. HR can focus on reducing the remaining attrition through effective retention strategies.

**4. Count Plot – Gender Distribution**

- The count plot displays the distribution of employees based on gender. Male employees represent a larger portion of the workforce compared to female employees. The organization maintains representation from both genders across various departments. This analysis supports diversity and workforce planning initiatives.

**5. Box Plot – Department-wise Monthly Income Distribution**

- The box plot compares monthly income across different departments. Employees in managerial and technical departments generally receive higher salaries, while salary variations exist within every department. Several outliers indicate employees earning significantly higher incomes than the average. This visualization helps identify salary distribution and pay consistency.

**6. Count Plot – Over Time vs Employee Attrition**

- The count plot compares employee attrition based on overtime status. Employees who frequently work overtime show a noticeably higher attrition rate than those who do not. This suggests that excessive workload may contribute to employee turnover. HR can use this insight to improve work-life balance and employee retention.

**7. Violin Plot – Monthly Income Distribution by Job Role**

- The violin plot illustrates salary distribution across different job roles. Managerial positions generally receive higher monthly incomes, while technical and sales roles show comparatively lower salary ranges. The spread of the distribution highlights differences in employee compensation across job categories. This helps HR evaluate salary structures by role.

**8. Scatter Plot – Age vs Monthly Income by Gender**

- The scatter plot shows the relationship between employee age and monthly income, categorized by gender. Monthly income generally increases with employee age and work experience. Male and female employees exhibit a similar income pattern without significant differences. This indicates that salary growth is primarily influenced by experience rather than gender.

**9. Correlation Heatmap**

- The correlation heatmap presents the relationship among numerical variables such as Age, Monthly Income, Total Working Years, and Years at Company. Monthly Income has a strong positive correlation with Total Working Years, indicating that experienced employees tend to earn higher salaries. Age also shows a moderate positive relationship with work experience. This visualization helps identify important factors influencing employee growth.

**10. Pair Plot**

- The pair plot visualizes relationships among Age, Monthly Income, Total Working Years, Years at Company, and Attrition. Employees with greater experience generally have higher salaries and longer organizational tenure. The distributions of employees with and without attrition overlap in most numerical features, indicating that attrition is influenced by multiple factors. This analysis provides a comprehensive understanding of employee characteristics.

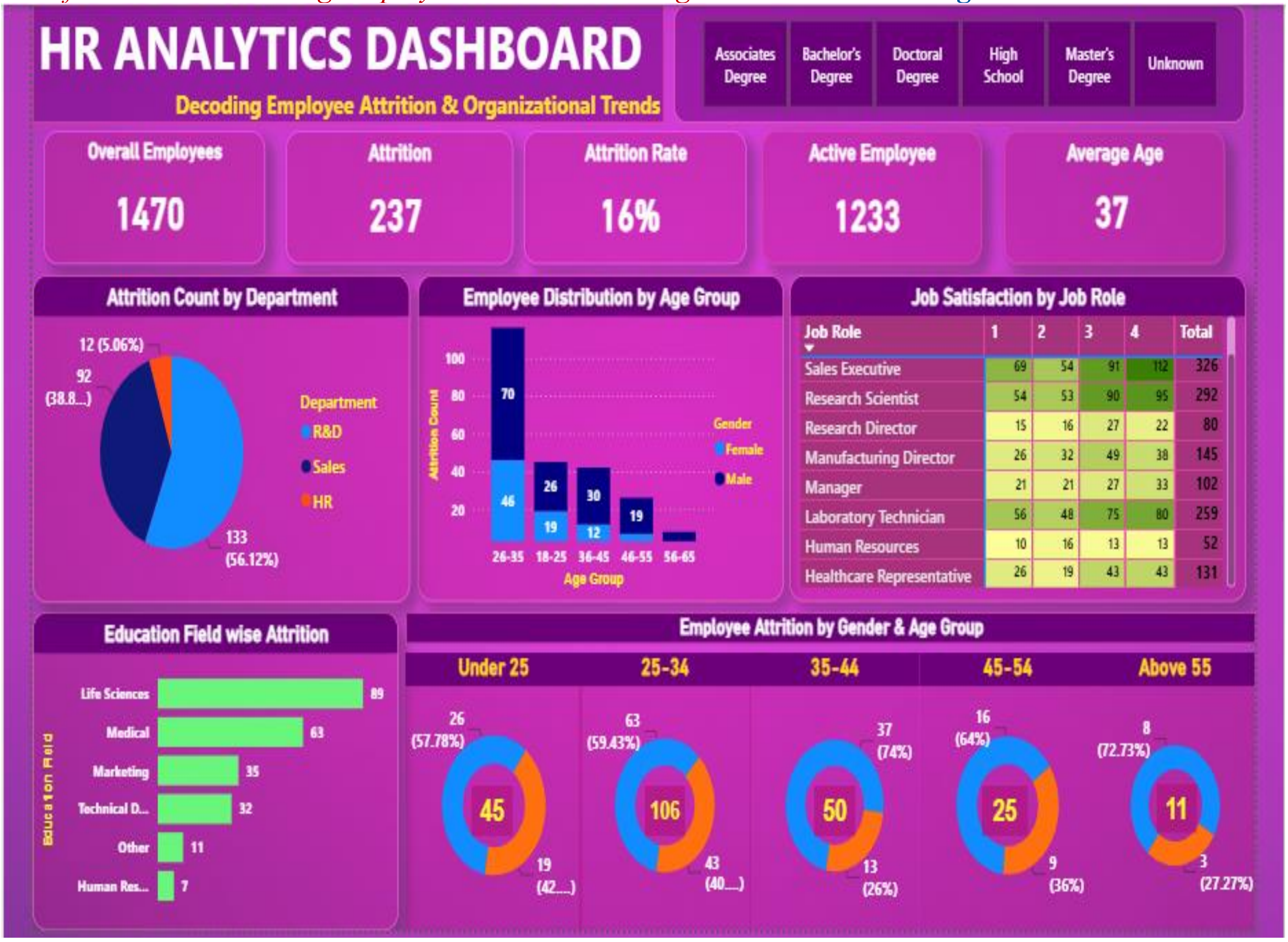
## POWER BI DESKTOP

## Data Integration & Dashboard Development in Power BI

- The cleaned HR Analytics dataset was exported from Python in **CSV** format and imported into **Power BI Desktop** for interactive dashboard development.
- Verified the data types of all columns and converted numerical fields into appropriate formats to ensure accurate calculations and visualizations.
- Created calculated measures using DAX for **Overall Employees, Attrition Count, Active Employees, Attrition Rate,** and **Average Age**.
- Developed relationships between employee demographic, job-related, and satisfaction attributes to generate meaningful HR insights.
- Designed an interactive dashboard using KPI Cards, Pie Chart, Clustered Column Chart, Bar Chart, Matrix Table, Donut Charts, and Slicers for dynamic analysis.

## HR Analytics Dashboard Overview

- The **HR Analytics** Dashboard was designed to provide a comprehensive overview of employee demographics, attrition trends, job satisfaction, and workforce performance. The dashboard enables HR professionals to analyze employee behavior through interactive visualizations and dynamic filters.
- To improve data analysis and visualization, additional calculated columns such as **Income Level, Age Group**, and **Age Band** were created during data transformation. The Age Band column categorizes employees into **Under 25, 25–34, 35–44, 45–54, and Above 55 groups**, making age-wise attrition analysis more meaningful.




## Dashboard Analysis & Key Findings

### KPI Cards

- The organization consists of **1,470 employees**, of which **237 employees have left** the company.
- The overall **Attrition Rate is 16%**, while **1,233 employees remain active**.
- The **Average Employee Age is 37 years**, indicating a workforce dominated by experienced professionals.

### Attrition Count by Department

- The **Research & Development (R&D)** department records the highest attrition with **133 employees**, followed by **Sales (92)** and **Human Resources (12)**.
- This indicates that employee turnover is more concentrated in departments with larger workforces.

**Employee Distribution by Age Group**

- Employees aged **26–35 years** represent the largest portion of the workforce.
- Employee count gradually decreases in higher age groups, indicating that the organization has a relatively young workforce.

**Job Satisfaction by Job Role**

- **Sales Executives** have the highest employee count among all job roles.
- Most job roles show satisfaction levels distributed across all four rating categories, indicating varying employee experiences within the organization.

**Education Field-wise Attrition**

- Employees from the **Life Sciences** education field show the highest attrition, followed by **Medical** and **Marketing** backgrounds.
- Human Resources education field records the lowest employee attrition.

**Employee Attrition by Gender & Age Group**

- Employees aged **25–34 years** experience the highest employee attrition.
- Male employees contribute slightly more to attrition than female employees across most age groups.
- Employees above **55 years** record the lowest attrition, indicating higher organizational stability among senior employees.

## Project Conclusion & Strategic HR Recommendations

## Project Conclusion:

- The HR Analytics project successfully analyzed employee demographics, job satisfaction, education background, and attrition patterns using Python and Power BI. The interactive dashboard revealed that the organization has **1,470 employees**, with **237 employees leaving the organization**, resulting in an overall **16% attrition rate**. The analysis identified that the **Research & Development** department experienced the highest employee attrition, while employees in the **26–35 years** age group and those from the **Life Sciences** education field showed higher turnover. Overall, the dashboard provides valuable insights that enable HR professionals to monitor workforce trends, improve employee engagement, and make informed strategic decisions.

## Strategic HR Recommendations:

- Develop targeted employee retention programs for the **Research & Development** and **Sales** departments, where attrition levels are comparatively higher.
- Strengthen career growth opportunities, employee recognition, and learning initiatives for employees in the **26–35 years** age group to improve long-term retention.
- Conduct regular employee satisfaction and engagement surveys to identify workplace concerns and resolve them proactively.
- Improve work-life balance by monitoring employee workload and encouraging healthy workplace practices to reduce employee turnover.
- Provide skill development, leadership training, and career progression programs to enhance employee motivation and organizational productivity.
- Use HR Analytics dashboards as a continuous decision-support tool for workforce planning, performance monitoring, and employee retention strategies.

## Future Enhancements & Scope of Improvement

- Integrate **Machine Learning** models to predict employee attrition before employees decide to leave the organization.
- Develop predictive models for employee performance and promotion eligibility using historical HR data.
- Integrate real-time HR databases with **Power BI Service** for continuous workforce monitoring and reporting.
- Build AI-based employee recommendation systems to support recruitment, retention, and workforce planning.
- Expand the dashboard by incorporating employee attendance, leave management, performance evaluation, and payroll analytics for comprehensive Human Resource decision-making.